

SOUTH ROSEMARY, NC, USA

WMO: 724008

Lat: 36.439N

Lon: 77.710W

Elev: 256

StdP: 14.56

Time Zone: -5.00 (NAE)

Period: 99-09

WBAN: 93781

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.5	24.8	1.2	5.9	26.6	5.7	7.4	29.9	20.4	40.5	18.4	41.6	5.0	0	0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	93.9	75.4	91.4	75.1	89.9	74.5	78.7	89.2	77.5	87.6	76.6	85.8	5.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.4	134.3	82.8	74.8	131.5	82.0	73.3	125.0	80.1	42.3	89.3	41.0	87.9	40.1	85.5	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	14.3	12.1	14.8	97.4	3.3	3.9	12.5	100.2	10.6	102.5	8.7	104.7	6.3	107.5
			12.3	81.2	2.9	1.6	10.2	82.3	8.5	83.3	6.9	84.2	4.8	85.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.6	41.4	43.5	51.2	60.0	67.6	75.7	78.6	78.4	72.0	61.3	52.8	43.9	(6)
	DBStd	15.42	11.11	8.38	9.09	8.36	6.82	5.37	4.41	4.65	5.63	8.22	8.74	8.97	(7)
	HDD50	954	318	215	97	11	0	0	0	0	0	8	73	232	(8)
	HDD65	3235	732	603	436	196	55	2	0	0	13	172	372	654	(9)
	CDD50	4834	53	33	134	311	546	771	888	881	660	357	156	44	(10)
	CDD65	1640	1	1	9	46	136	322	423	416	223	56	6	1	(11)
	CDH74	14537	3	3	96	487	1189	2948	3990	3818	1575	376	47	5	(12)
	CDH80	5632	0	0	9	131	340	1201	1714	1660	488	88	1	0	(13)
(14) Wind	WSAvg	5.0	5.8	5.8	6.0	6.0	5.0	4.5	4.0	3.7	4.5	4.5	4.8	5.4	(14)
(15) Precipitation	PrecAvg	43.3	3.3	2.5	4.2	3.0	3.3	4.1	4.8	4.4	4.4	3.3	3.0	3.3	(15)
	PrecMax	56.5	6.1	5.4	8.1	7.4	5.9	10.3	8.1	10.7	18.9	6.8	8.1	6.1	(16)
	PrecMin	34.2	1.1	0.5	0.9	1.2	0.9	1.5	2.0	0.6	0.1	0.0	0.6	0.9	(17)
	PrecStd	6.3	1.3	1.2	1.7	1.4	1.5	2.4	1.6	2.3	3.8	1.7	1.9	1.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.5	73.0	81.4	88.5	90.1	97.0	96.0	98.6	91.5	87.6	79.0	73.0	(19)
		MCWB	61.6	59.2	63.1	67.1	71.9	73.6	77.2	77.1	72.4	72.9	65.0	61.2	(20)
	2%	DB	68.8	67.7	76.8	83.6	87.2	92.5	93.2	94.4	88.5	81.9	74.8	67.9	(21)
		MCWB	60.5	55.9	61.3	63.6	70.2	73.5	76.5	76.3	71.1	68.5	63.1	61.6	(22)
	5%	DB	64.0	63.6	72.1	80.6	83.7	89.9	91.0	91.2	84.8	78.6	71.9	63.3	(23)
		MCWB	57.6	54.0	58.7	63.9	69.0	73.4	76.3	75.5	70.6	66.2	62.1	57.7	(24)
	10%	DB	59.2	59.5	67.5	75.1	81.2	87.4	89.2	88.9	82.1	74.1	67.7	59.2	(25)
		MCWB	52.9	49.8	56.9	61.6	67.5	72.8	75.5	75.1	70.7	65.7	60.5	52.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.9	62.4	65.8	69.5	74.8	77.9	81.1	80.9	77.3	74.4	69.9	65.4	(27)
		MCDB	69.5	67.9	75.0	82.3	85.8	88.8	91.8	91.0	84.3	81.1	72.9	69.4	(28)
	2%	WB	61.4	59.5	63.6	67.7	72.3	76.6	79.0	78.9	75.5	71.9	66.3	62.6	(29)
		MCDB	67.2	65.0	70.9	80.0	82.8	86.8	89.2	89.9	82.1	77.9	71.3	66.0	(30)
	5%	WB	58.6	55.8	61.8	65.5	70.5	75.5	77.7	77.6	74.3	69.8	64.0	58.9	(31)
		MCDB	64.0	63.0	70.3	75.3	80.5	84.9	87.8	88.0	80.2	75.5	68.5	63.5	(32)
	10%	WB	54.2	50.8	58.9	63.7	68.9	74.5	76.7	76.5	72.9	67.3	61.4	53.9	(33)
		MCDB	58.0	58.3	66.4	71.5	77.9	83.1	86.1	85.6	78.5	72.7	67.3	58.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	17.8	19.7	21.3	21.5	20.4	18.8	17.9	18.1	18.1	19.6	20.3	18.6	(35)
		MCDBR	22.2	26.1	26.3	26.7	23.3	22.1	20.1	21.6	21.2	22.7	24.0	24.0	(36)
	5% WB	MCWBR	16.0	17.7	14.2	11.9	10.4	7.9	7.5	7.5	8.4	10.8	14.1	18.4	(37)
		MCWBR	21.1	22.6	23.3	22.9	21.0	18.0	18.2	18.6	16.0	17.4	19.3	21.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.302	0.317	0.347	0.391	0.438	0.487	0.515	0.514	0.404	0.358	0.320	0.304	(40)
		taud	2.527	2.485	2.433	2.315	2.210	2.122	2.063	2.052	2.356	2.471	2.532	2.560	(41)
	Ebn at Noon		285	293	292	283	270	256	247	245	270	274	276	277	(42)
		Edh at Noon	25	30	34	40	45	49	52	52	36	30	25	23	(43)
(44) All-Sky Solar Radiation	RadAvg		783	1009	1322	1682	1875	2023	1913	1733	1417	1137	864	687	(44)
	RadStd		60	118	103	132	147	130	96	95	123	143	70	57	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (66 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+279	+191

Nomenclature: See separate page